



CMPT 413/713: Natural Language Processing

Project tips and analyzing your results

Fall 2021
2021-11-20

Project Details

- Project proposal feedback out this evening
- Project milestone due Thursday 11/18
- Project “Poster” presentation on last day of this class (Tuesday 12/7) - tentative plan
 - By 12/5: Short video (3 min) summarizing your project
 - Per group zoom link that others can use to visit your presentations
 - Use slides/material prepared for video
- Project final report due last day of all classes (Wednesday 12/8)

Project Milestone

- Project Milestone due **Thursday 11/18**
- **PDF (3-6 pages)** in the style of a conference (e.g. ACL/EMNLP) submission
 - <https://2020.emnlp.org/files/emnlp2020-templates.zip>
- Milestone should include:
 - Title and Abstract - motivate the problem, describe your goals, and highlight your findings
 - Approach - details on your main approach and baselines. Be specific. Make clear what part is original, what code you are writing yourself, what code you are using
 - Experiment - describe dataset, evaluation metrics, what experiments you plan to run, any results you have so far. Also provide training details, training times, etc.
 - Future Work - what is your plan for the rest of the project
 - Reference - provide references using BibTex
- Milestone will be graded based on progress and writing quality

Project Milestone

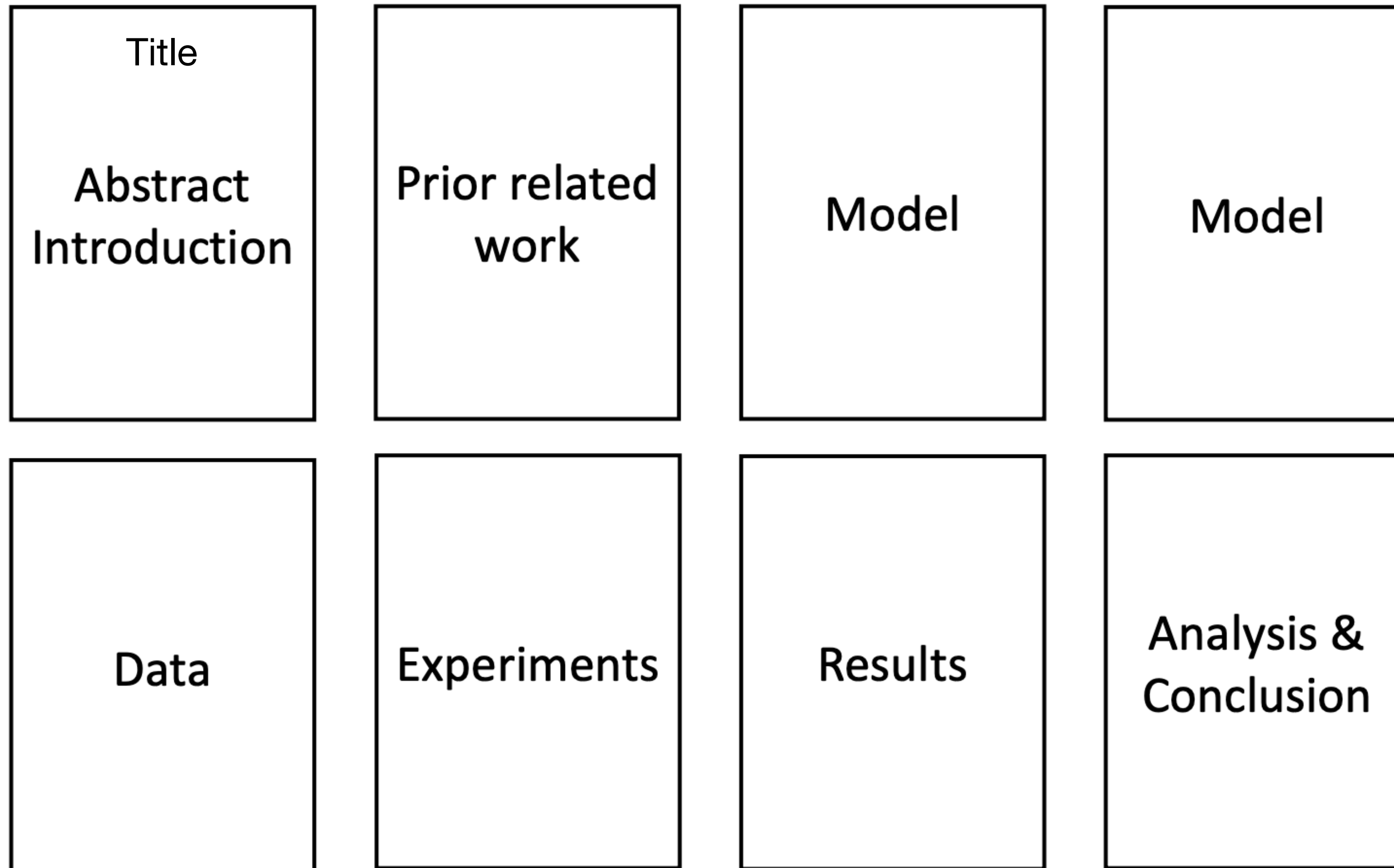
- Build on what you have done for the proposal
- Include progress and initial results
- Include dataset examples and statistics
- Include figures for your model
- Include references
- Make it clear what you are implementing vs what part you are building on top of existing libraries / codebases / homework

Final project report

8 pages not including references

Model or what approach you are taking

Rough page layout:
your report can have
different number of
pages for each
section

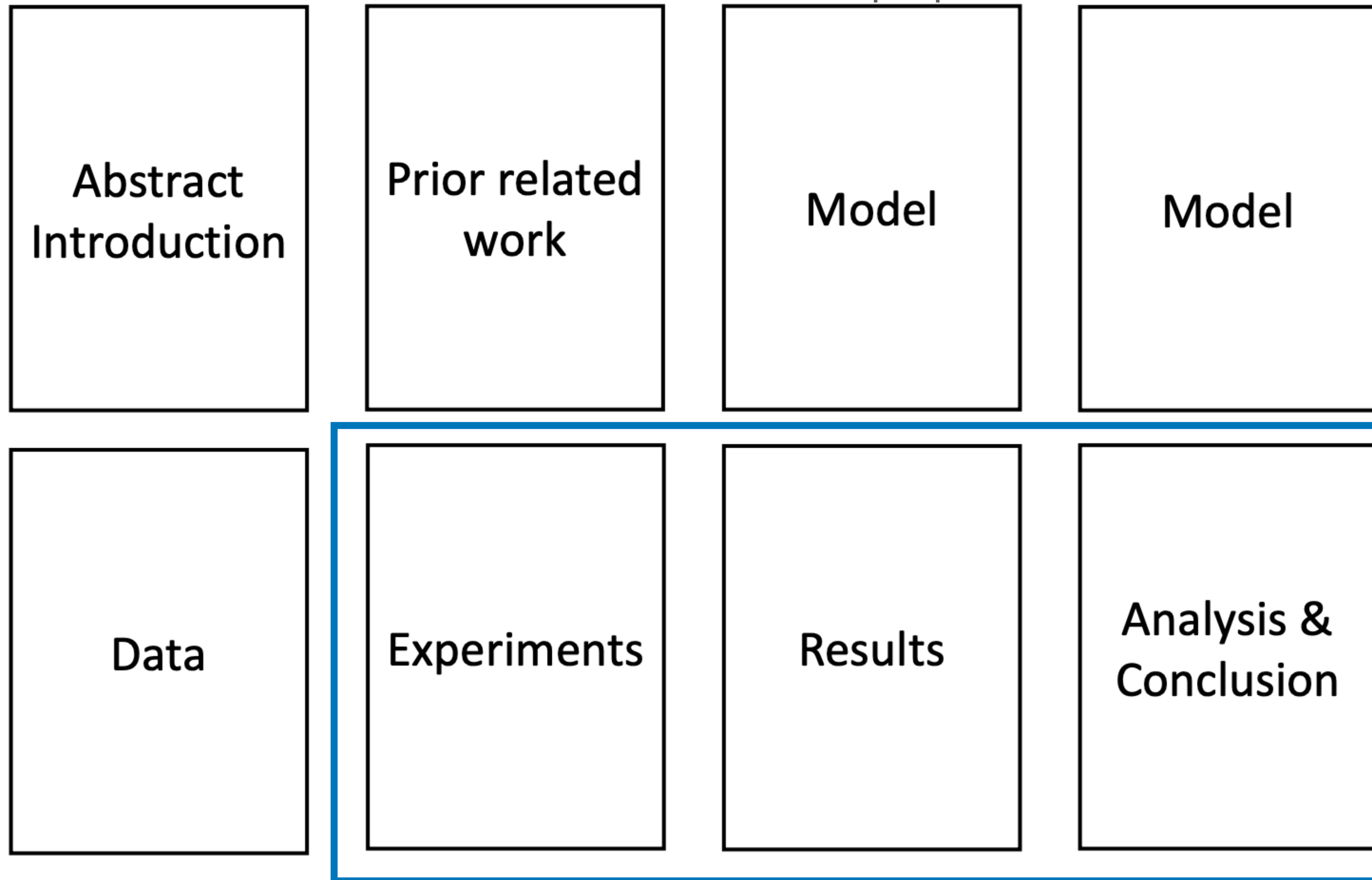


Details of what goes into a paper

1. **Intro**: Tell the full story of your paper at a high-level.
2. **Related Work**: Contextualize your work and provide insights into major relevant themes of the literature as a whole. Use each paper (or theme) as a chance to articulate what is special about your paper.
3. **Data**: Likely to be very detailed if the datasets are new or unfamiliar to the community, or if familiar datasets are being used in new ways.
4. **Model**: Flesh out your own approach, perhaps amplifying themes from the 'Related work' section.
5. **Experiments**: The experimental approach, including descriptions of metrics, baseline models, etc. Details about hyperparameters, optimization choices, etc., are probably best given in appendices, unless they are central to the arguments.
6. **Results**: A no-nonsense report of what happened.
7. **Analysis**: Discussion of what the results mean, what they don't mean, where they can be improved, etc. These sections vary a lot depending on the nature of the paper.
8. (For papers reporting on experiments with multiple datasets, it can be good to repeat Methods/Results/Analysis in separate (sub)sections for each dataset.)
9. **Conclusion**: Quickly summarize what the paper did, and then chart out possible future directions that anyone might pursue.

Milestone report as a draft for final project report

We expect you to have around 3-6 pages, can be more (but < 8) if you have done lots of work since the proposal



Not expected to be complete:

- preliminary experiments
- difficulties
- what is still to be done

Have **statistics/analysis** of your data, and show **examples!**

Include References

Why a page limit?

- Good writing is concise and clear
- More words does not mean more information or higher quality
- Trim words you don't need.
- But I have a lot of stuff to say!
 - References don't count toward the page limit
 - You can have appendices if you really want to share (the appendices will not be graded)

Proposal Observations

- Use the correct format
- References
 - Use bibtex and \cite commands
 - Make sure that you cite the papers that introduced the method you are using (word2vec, BERT, LSTM, GRU, Transformer, etc).
 - Make sure that the references are properly included
- English
 - Use complete sentences
 - Proofread your report
 - Have friend (native/fluent English speaker) proofread your report

Proposal Observations

- Summarize relevant work
 - Summarize what did they do? What was the key findings?
 - (this requires larger changes than just replacing “we” with “they”)
 - Avoid copying text from another paper verbatim (if you copy, it is plagiarism).
 - Describe the approach in your own words. Use different sentence structures.
- Describe your problem
 - What is your **input**? What is your **output**?
- Describe how you will use a particular method for **your task**
 - It’s great that you are using RNN, LSTM, GRU, and attention, and know the differences between the models.
 - But how will you actually use it for your task?

What is your project about?

Your report should be different from what we cover in Lecture and may be different from actual research papers

- Lecture
 - Teach you about specific models (FFN,RNN/Transformers)
 - Focus is on the inner workings of the model and how it is hooked up
- Research papers
 - Invent a new model / loss / training
 - Focus is mostly on whatever is new or being studied (method or experiments)
- What should you focus on describing?
 - What is your project about?
 - Think about what do you need to explain

Timeline

- Make sure to allocate time for training and debugging the training of your model.
- In addition, you should allocate time for setting up the experiments (train classifier with some set of hyperparameters), evaluate the classifier, and analyzing the results.
- Plan for report writing and video making

Tips for good final projects

- Have a clear, well-defined hypothesis to be tested
(++ novel/creative hypothesis)
- Conclusions and results should teach the reader something
- Meaningful **tables, plots** to display the key results

++ nice visualizations or interactive demos

++ novel/impressive engineering feat

++ good results

What to avoid

- All experiments run with prepackaged source - no extra code written for model/data processing
- Just ran model once or twice on the data and reported results (not much hyperparameter search done)
- A few standard graphs: loss curves, accuracy, without any analysis
- Results/Conclusion don't say much besides that it didn't work
 - Even if results are negative, **you should analyze them!**

Remember:
Include
analysis!

What makes for a good paper? (example paper)

A Thorough Examination of the CNN/Daily Mail Reading Comprehension Task
<https://aclanthology.org/P16-1223.pdf> (Chen et al, ACL 2016, Outstanding Paper)

Task definition

Example with input and output

Text describing the problem

an example⁴: it consists of a passage p , a question q and an answer a , where the passage is a news article, the question is a cloze-style task, in which one of the article's bullet points has had one entity replaced by a placeholder, and the answer is this questioned entity. The goal is to infer the missing entity (answer a) from all the possible entities which appear in the passage. A news article is usually

Passage

(@entity4) if you feel a ripple in the force today , it may be the news that the official @entity6 is getting its first gay character . according to the sci-fi website @entity9 , the upcoming novel " @entity11 " will feature a capable but flawed @entity13 official named @entity14 who " also happens to be a lesbian . " the character is the first gay figure in the official @entity6 -- the movies , television shows , comics and books approved by @entity6 franchise owner @entity22 -- according to @entity24 , editor of " @entity6 " books at @entity28 imprint @entity26 .

Question

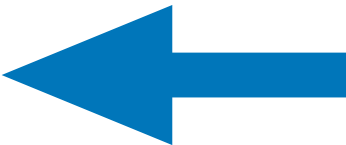
characters in " @placeholder " movies have gradually become more diverse

Answer

@entity6

Taken from the Reading Comprehension dataset introduced by Herman et al, 2015

Goal: identify entity that is goes where @placeholder goes



A Thorough Examination of the CNN/Daily Mail Reading Comprehension Task
<https://aclanthology.org/P16-1223.pdf> (Chen et al, ACL 2016, Outstanding Paper)

Task definition

- Dataset is automatically generated
- Use Google NLP pipeline and get entities and coreference chains
- Cloze-style (fill in the blank) questions generated by taking sentence from passage and replacing entity reference with @placeholder
- Data is anonymized (entities are just @entity#)

Example with input and output

Passage

(@entity4) if you feel a ripple in the force today , it may be the news that the official @entity6 is getting its first gay character . according to the sci-fi website @entity9 , the upcoming novel " @entity11 " will feature a capable but flawed @entity13 official named @entity14 who " also happens to be a lesbian . " the character is the first gay figure in the official @entity6 -- the movies , television shows , comics and books approved by @entity6 franchise owner @entity22 -- according to @entity24 , editor of " @entity6 " books at @entity28 imprint @entity26 .

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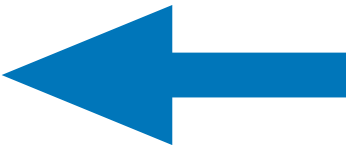
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Approach

Two approaches presented

In this section, we describe two systems we implemented – a conventional entity-centric classifier and an end-to-end neural network. While Hermann

Features are described for entity-centric classifier

Describe problem again
(with math symbols)

Given the (passage, question, answer) triple (p, q, a) , $p = \{p_1, \dots, p_m\}$ and $q = \{q_1, \dots, q_l\}$ are sequences of tokens for the passage and question sentence, with q containing exactly one “@placeholder” token. The goal is to infer the correct entity $a \in p \cap E$ that the placeholder corresponds to, where E is the set of all abstract entity markers. Note that the correct answer entity must appear in the passage p .

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Model description

- Describe key parts including how it is hooked out to input and how prediction is made

Encoding: First, all the words are mapped to d -dimensional vectors via an embedding matrix $E \in \mathbb{R}^{d \times |\mathcal{V}|}$; therefore we have p : $\mathbf{p}_1, \dots, \mathbf{p}_m \in \mathbb{R}^d$ and q : $\mathbf{q}_1, \dots, \mathbf{q}_l \in \mathbb{R}^d$.

Prediction: Using the *output* vector $\mathbf{o}$, the system outputs the most likely answer using:

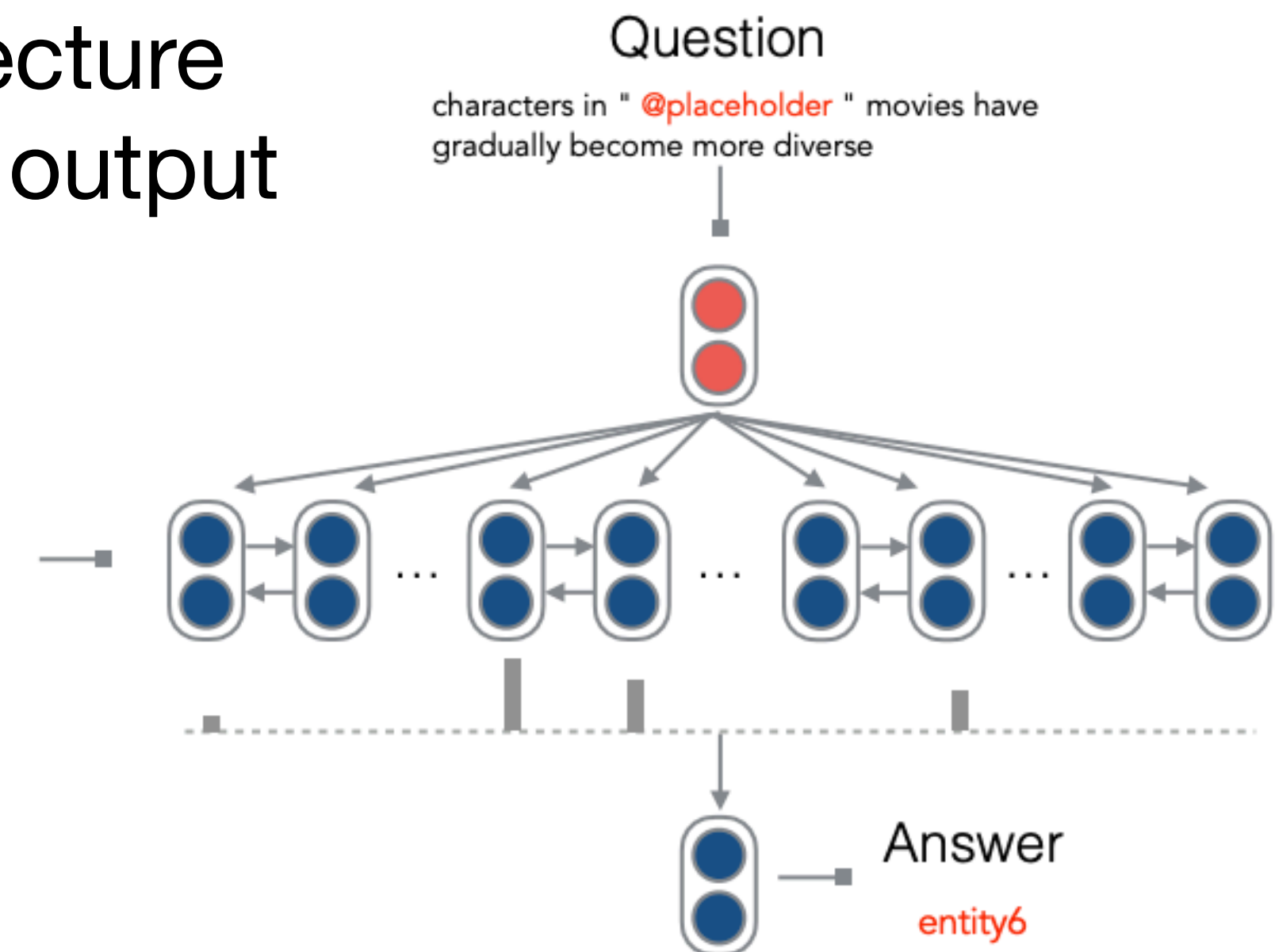
$$a = \arg \max_{a \in p \cap E} W_a^T \mathbf{o} \quad (4)$$

Symbols are connected

Model architecture with input and output

Passage

(@entity4) if you feel a ripple in the force today , it may be the news that the official @entity6 is getting its first gay character . according to the sci-fi website @entity9 , the upcoming novel " @entity11 " will feature a capable but flawed @entity13 official named @entity14 who " also happens to be a lesbian . " the character is the first gay figure in the official @entity6 -- the movies , television shows , comics and books approved by @entity6 franchise owner @entity22 -- according to @entity24 , editor of " @entity6 " books at @entity28 imprint @entity26 .



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Provide implementation and training details

Training Details (for entity-centric model)

- What external software / code was used
- Why was it used?

For training our conventional classifier, we use the implementation of *LambdaMART* (Wu et al., 2010) in the RankLib package.⁵ We use this ranking algorithm since our problem is naturally a ranking problem and forests of boosted decision trees have been very successful lately (as seen, e.g., in many recent Kaggle competitions). We do not use all the features of *LambdaMART* since we are only scoring 1/0 loss on the first ranked proposal, rather than using an IR-style metric to score ranked results. We use Stanford's neural network dependency parser (Chen and Manning, 2014) to parse all our document and question text, and all other features can be extracted without additional tools.

Provide implementation and training details

Training Details (for NN model)

- Vocabulary size, UNK handling
- Word embeddings
- Parameter initialization
- Dimensions (LSTM hidden state, word embeddings)
- Optimizer: type, learning rate, mini-batch size, dropout
- Number of epochs trained, how the “best” model was selected
- Compute resource uses (GPU type, runtime)

For training our neural networks, we only keep the most frequent $|\mathcal{V}| = 50\text{k}$ words (including entity and placeholder markers), and map all other words to an *<unk>* token. We choose word embedding size $d = 100$, and use the 100-dimensional pre-trained *GloVe* word embeddings (Pennington et al., 2014) for initialization. The attention and output parameters are initialized from a uniform distribution between $(-0.01, 0.01)$, and the LSTM weights are initialized from a Gaussian distribution $\mathcal{N}(0, 0.1)$.

We use hidden size $h = 128$ for *CNN* and 256 for *Daily Mail*. Optimization is carried out using

vanilla stochastic gradient descent (SGD), with a fixed learning rate of 0.1. We sort all the examples by the length of its passage, and randomly sample a mini-batch of size 32 for each update. We also apply dropout with probability 0.2 to the embedding layer and gradient clipping when the norm of gradients exceeds 10.

All of our models are run on a single GPU (GeForce GTX TITAN X), with roughly a runtime of 6 hours per epoch for *CNN*, and 15 hours per epoch for *Daily Mail*. We run all the models up to 30 epochs and select the model that achieves the best accuracy on the development set.

Provide implementation and training details

Organize hyper
parameters in a table
if appropriate

Hyperparam	BERT	RoBERTa	ALBERT
Epochs	3, 10, 20	3	3
Learning rate	$1e-5 - 5e-5$	$1e-5 - 3e-5$	$1e-5 - 3e-5$
Learning rate schedule	warmup-linear	warmup-linear	warmup-linear
Warmup ratio	0.1	0.1	0.1
Batch size	16	16	16
Adam ϵ	$1e-6$	$1e-6$	$1e-6$
Adam β_1	0.9	0.9	0.9
Adam β_2	0.999	0.98	0.999
Adam bias correction	{True, False}	{True, False}	{True, False}
Dropout	0.1	0.1	–
Weight decay	0.01	0.1	–
Clipping gradient norm	1.0	–	1.0
Number of random seeds	25	25	25

On the stability of fine-tuning BERT: Misconceptions, explanations, and strong baselines
<https://arxiv.org/pdf/2006.04884v3.pdf> (Mosbach et al, ICLR 2021)

Provide data statistics

- Statistics about what goes into train/val/test
- NLP specific statistics (number of tokens, sentences, unique words, etc)

	CNN	Daily Mail
# Train	380,298	879,450
# Dev	3,924	64,835
# Test	3,198	53,182
Passage: avg. tokens	761.8	813.1
Passage: avg. sentences	32.3	28.9
Question: avg. tokens	12.5	14.3
Avg. # entities	26.2	26.2

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Summarize your results in tables

Describe key findings
in the text

(Hermann
et al., 2015)

(Hill et al., 2016)

Ranking classifier with
decision forests
Careful feature selection

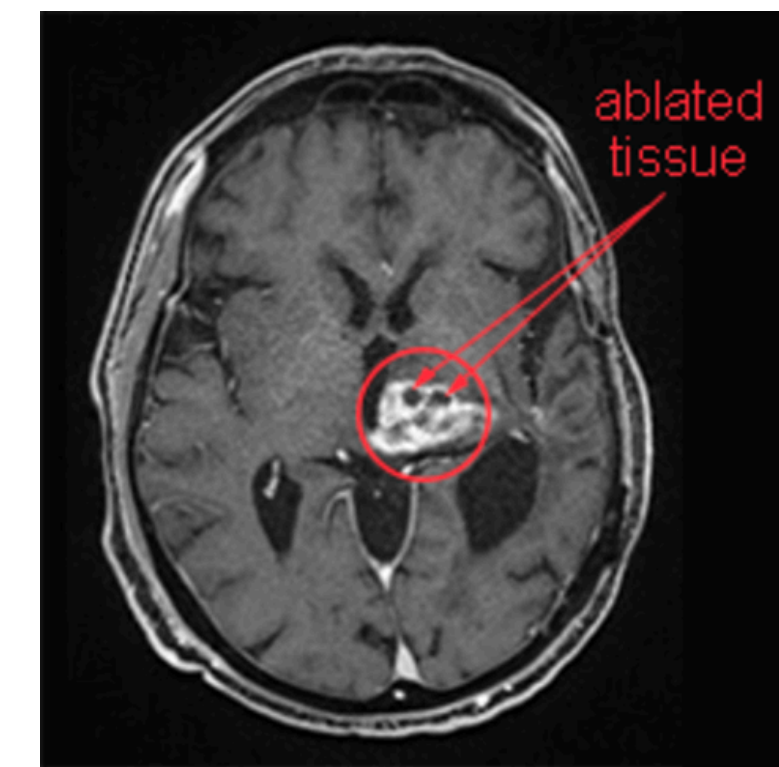
Model	CNN		Daily Mail	
	Dev	Test	Dev	Test
Frame-semantic model [†]	36.3	40.2	35.5	35.5
Word distance model [†]	50.5	50.9	56.4	55.5
Deep LSTM Reader [†]	55.0	57.0	63.3	62.2
Attentive Reader [†]	61.6	63.0	70.5	69.0
Impatient Reader [†]	61.8	63.8	69.0	68.0
MemNNs (window memory) [‡]	58.0	60.6	N/A	N/A
MemNNs (window memory + self-sup.) [‡]	63.4	66.8	N/A	N/A
MemNNs (ensemble) [‡]	66.2*	69.4*	N/A	N/A
Ours: Classifier	67.1	67.9	69.1	68.3
Ours: Neural net	72.4	72.4	76.9	75.8

NN Model is similar to Attentive Reader, compared to Attentive Reader

- Use bilinear attention instead of tanh layer
- Simpler model (only use weights contextual embeddings, don't classify over all vocabulary but only words that appear in passage)

Conduct ablation studies

to ablate = to remove



- If classical classifier (Naive Bayes, logistic regression), do feature ablation to see how important is each feature
- If neural network with different components, ablate components.
- If loss is combination of terms, ablate loss terms.

Features	Accuracy
Full model	67.1
– whether e is in the passage	67.1
– whether e is in the question	67.0
– frequency of e	63.7
– position of e	65.9
– n -gram match	60.5
– word distance	65.4
– sentence co-occurrence	66.0
– dependency parse match	65.6

A Thorough Examination of the CNN/Daily Mail Reading Comprehension Task
<https://aclanthology.org/P16-1223.pdf> (Chen et al, ACL 2016, Outstanding Paper)

Perform data analysis

- Manual analysis of 100 samples from val (dev) split
- Categorize into 6 different types (based on relationship of passage/question with answer)

No.	Category	(%)
1	Exact match	13
2	Paraphrasing	41
3	Partial clue	19
4	Multiple sentences	2
5	Coreference errors	8
6	Ambiguous / hard	17

- Answerable categories:
 - Exact match
 - Paraphrasing
 - Partial clue
 - Multiple sentences
- Unanswerable categories:
 - Coreference error
 - The dataset was automatically generated
 - Ambiguous or very hard
 - Humans are not likely to be able to answer these

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<https://aclanthology.org/P16-1223.pdf> (Chen et al, ACL 2016, Outstanding Paper)

Concrete examples for each category

Category	Question	Passage
Exact Match	<i>it 's clear @entity0 is leaning toward @placeholder</i> , says an expert who monitors @entity0	... @entity116 , who follows @entity0 's operations and propaganda closely , recently told @entity3 , <i>it 's clear @entity0 is leaning toward @entity60</i> in terms of doctrine , ideology and an emphasis on holding territory after operations
Para-phrase	@placeholder says he understands why @entity0 wo n't play at his tournament	... @entity0 called me personally to let me know that he would n't be playing here at @entity23 , " @entity3 said on his @entity21 event 's website
Partial clue	a tv movie based on @entity2 's book @placeholder casts a @entity76 actor as @entity5	... to @entity12 @entity2 professed that his @entity11 is not a religious book
Multiple sent.	he 's doing a his - and - her duet all by himself , @entity6 said of @placeholder	... we got some groundbreaking performances , here too , tonight , @entity6 said . we got @entity17 , who will be doing some musical performances . he 's doing a his - and - her duet all by himself
Coref. Error	rapper @placeholder " disgusted , " cancels upcoming show for @entity280	... with hip - hop star @entity246 saying on @entity247 that he was canceling an upcoming show for the @entity249 (but @entity249 = @entity280 = SAEs)
Hard	pilot error and snow were reasons stated for @placeholder plane crash	... a small aircraft carrying @entity5 , @entity6 and @entity7 the @entity12 @entity3 crashed a few miles from @entity9 , near @entity10 , @entity11

Analysis leads to insights

- How many can the two models get correct for each category?
- Neural network is at close to optimal performance on this dataset!

(%)	Category	Classifier	Neural net
13	Exact match	13 (100.0%)	13 (100.0%)
41	Paraphrasing	32 (78.1%)	39 (95.1%)
19	Partial clue	14 (73.7%)	17 (89.5%)
2	Multiple sentences	1 (50.0%)	1 (50.0%)
8	Coreference errors	4 (50.0%)	3 (37.5%)
17	Ambiguous / hard	2 (11.8%)	1 (5.9%)
	All	66 (66.0%)	74 (74.0%)

Analyzing your results

Analyzing your results

- Understanding the output of your model
 - Look at the output and examine what it is getting right and what it is getting wrong
 - Provide examples
 - Conduct data / error analysis
 - Visualize
- Characterizing your model performance
 - Compare against other models
 - Compare against variations
 - Ablation studies

Provide qualitative examples of your model output

- Simple tip: color code **correct** and **incorrect** output

English-German translations

src	Orlando Bloom and Miranda Kerr still love each other
ref	Orlando Bloom und <i>Miranda Kerr</i> lieben sich noch immer
best	Orlando Bloom und <i>Miranda Kerr</i> lieben einander noch immer .
base	Orlando Bloom und Lucas Miranda lieben einander noch immer .
src	" We ' re pleased the FAA recognizes that an enjoyable passenger experience is not incompatible with safety and security , " said Roger Dow , CEO of the U.S. Travel Association .
ref	" Wir freuen uns , dass die FAA erkennt , dass ein angenehmes Passagiererlebnis nicht im Widerspruch zur Sicherheit steht " , sagte <i>Roger Dow</i> , CEO der U.S. Travel Association .
best	" Wir freuen uns , dass die FAA anerkennt , dass ein angenehmes ist nicht mit Sicherheit und Sicherheit <i>unvereinbar</i> ist " , sagte <i>Roger Dow</i> , CEO der US - die .
base	" Wir freuen uns über die <unk> , dass ein <unk> <unk> mit Sicherheit nicht vereinbar ist mit Sicherheit und Sicherheit " , sagte <i>Roger Cameron</i> , CEO der US - <unk> .

Conduct data and error analysis

How to perform data and error analysis?

How to start?

- Take a manual subsample of the data
- Manually group and categorize them (like the paper we looked at)
 - You will need to figure out what these groups are!
- Compute some statistics
 - Overall percentage of categories
 - What is the performance on each category?

Conduct data and error analysis

How to perform data and error analysis?

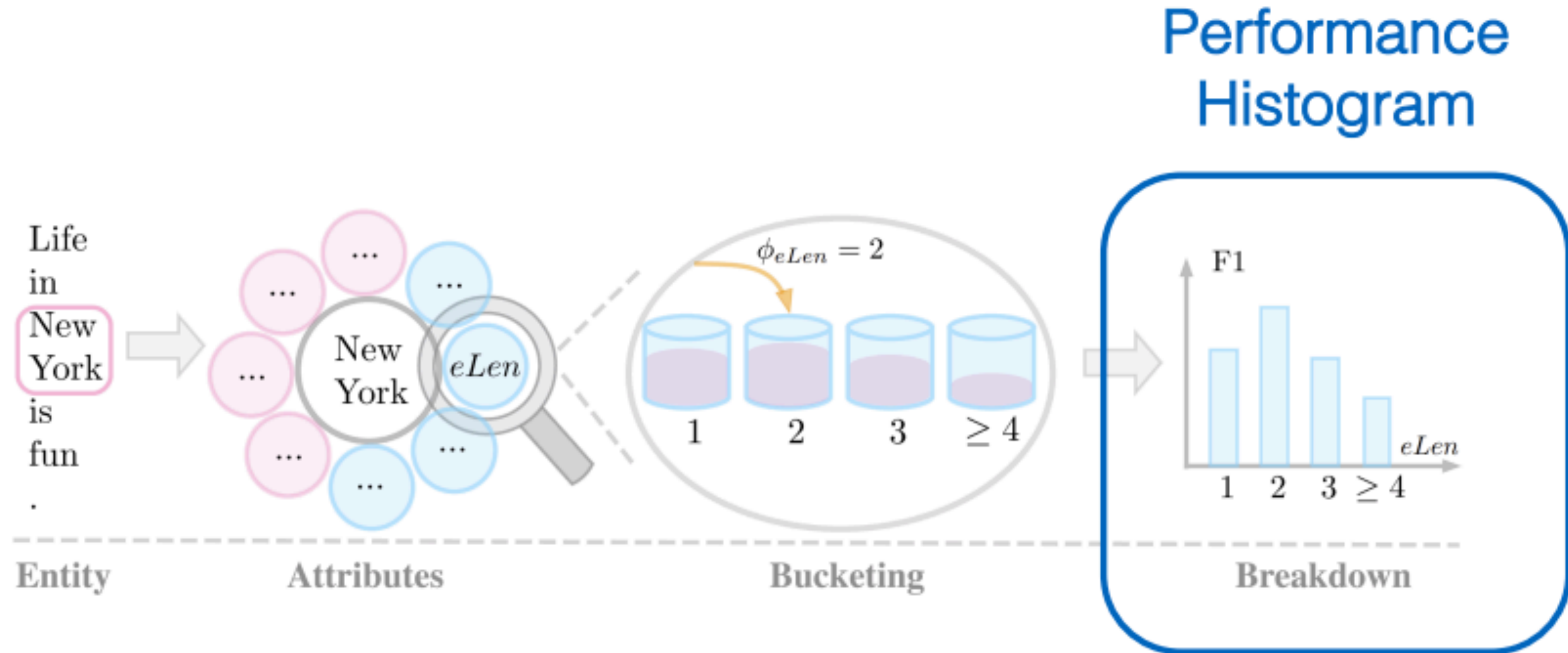
General recipe

- Partition the **performance** of the validation set into different **interpretable grouped** based on **pre-defined attributes**.
- Define attributes
- Group test samples
- Breakdown of performance

Defining attributes

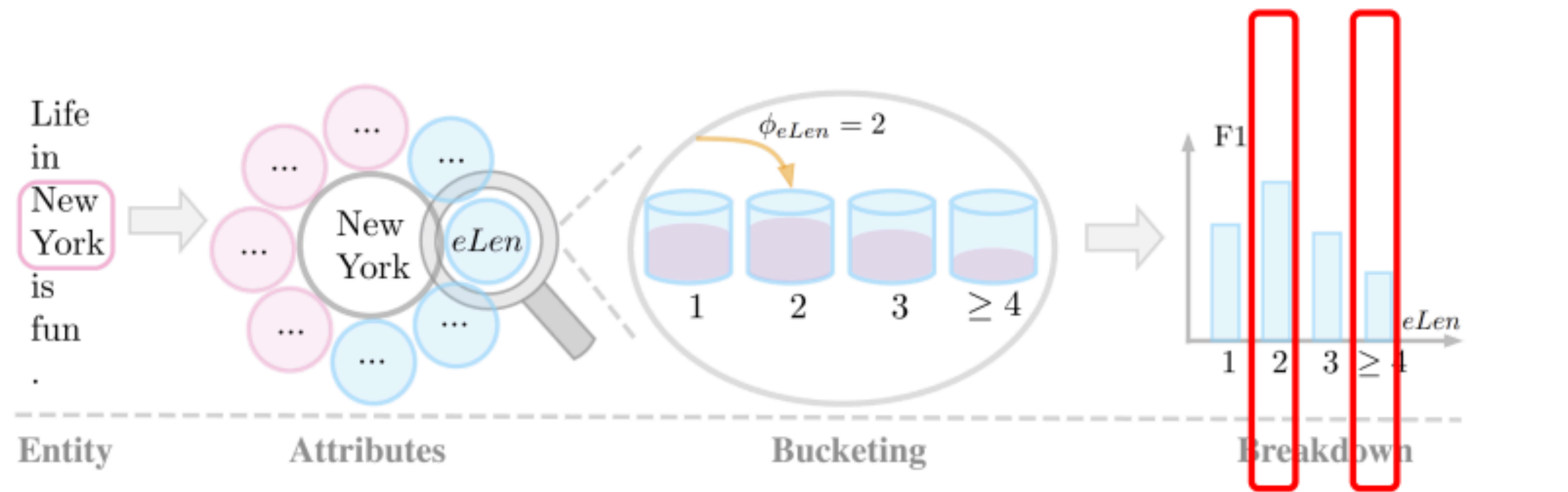
- Different tasks could have different attributes
- Token-level, span-level, sentence-level
 - Token-level: part-of-speech tag
 - Span-level: span length
 - Sentence-level: sentence length

Example: breakdown performance by entity length

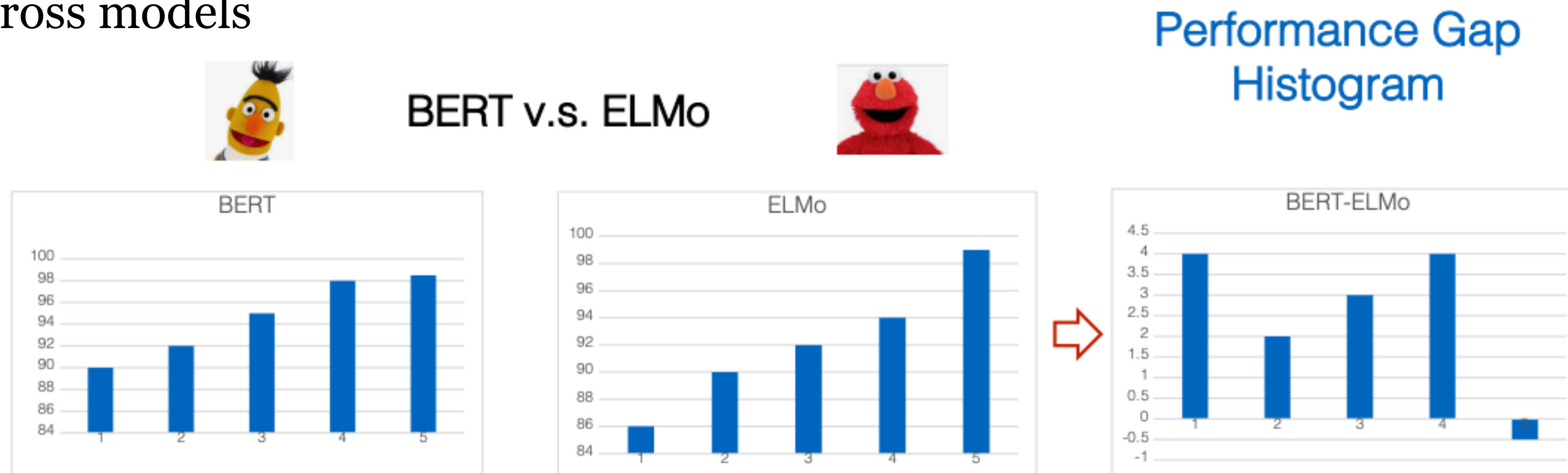


Performance histogram

- Compare across attributes

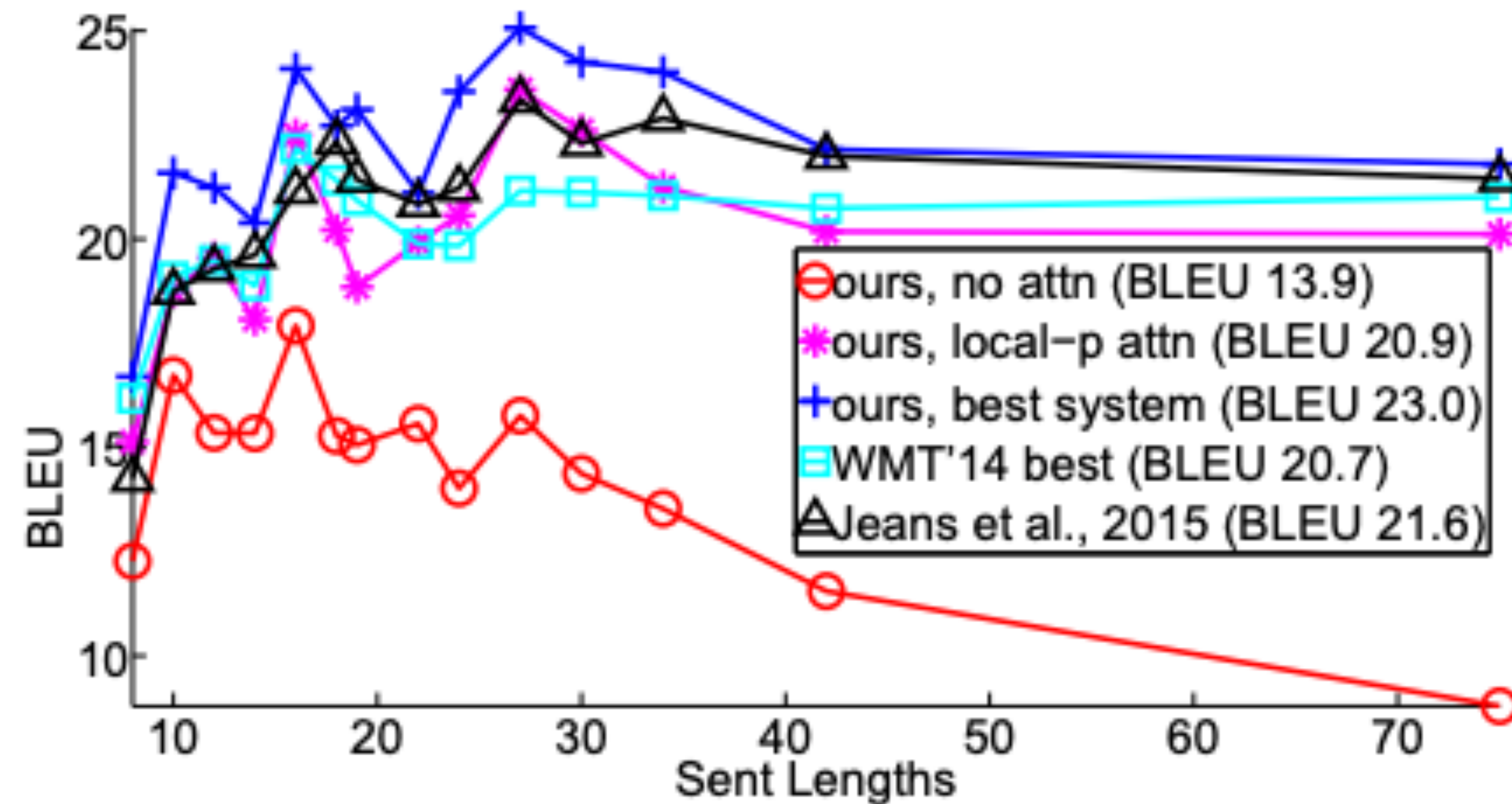


- Compare across models



Comparison of model performance by sentence length

- Attribute: sentence length



Integrating unit tests

- Small careful test sets sound like... unit test suites, but for neural networks!
- *Minimum functionality tests*: small test sets that target a specific behavior.

Test case	Expected	Predicted	Pass?
A Testing Negation with <i>MFT</i> Labels: negative, positive, neutral			
Template: I {NEGATION} {POS_VERB} the {THING}.			
I can't say I recommend the food.	neg	pos	X
I didn't love the flight.	neg	neutral	X
...			
Failure rate = 76.4%			

- Ribeiro et al., 2020 showed **ML engineers working on a sentiment analysis product** an interface with categories of linguistic capabilities and types of tests.
 - The engineers found a bunch of bugs (categories of high error) through this method!

Beyond Accuracy: Behavioral Testing of NLP models with CheckList
<https://arxiv.org/pdf/2005.04118.pdf> (Ribeiro et al, ACL 2020)

Slide credit: John Hewitt

Comparisons

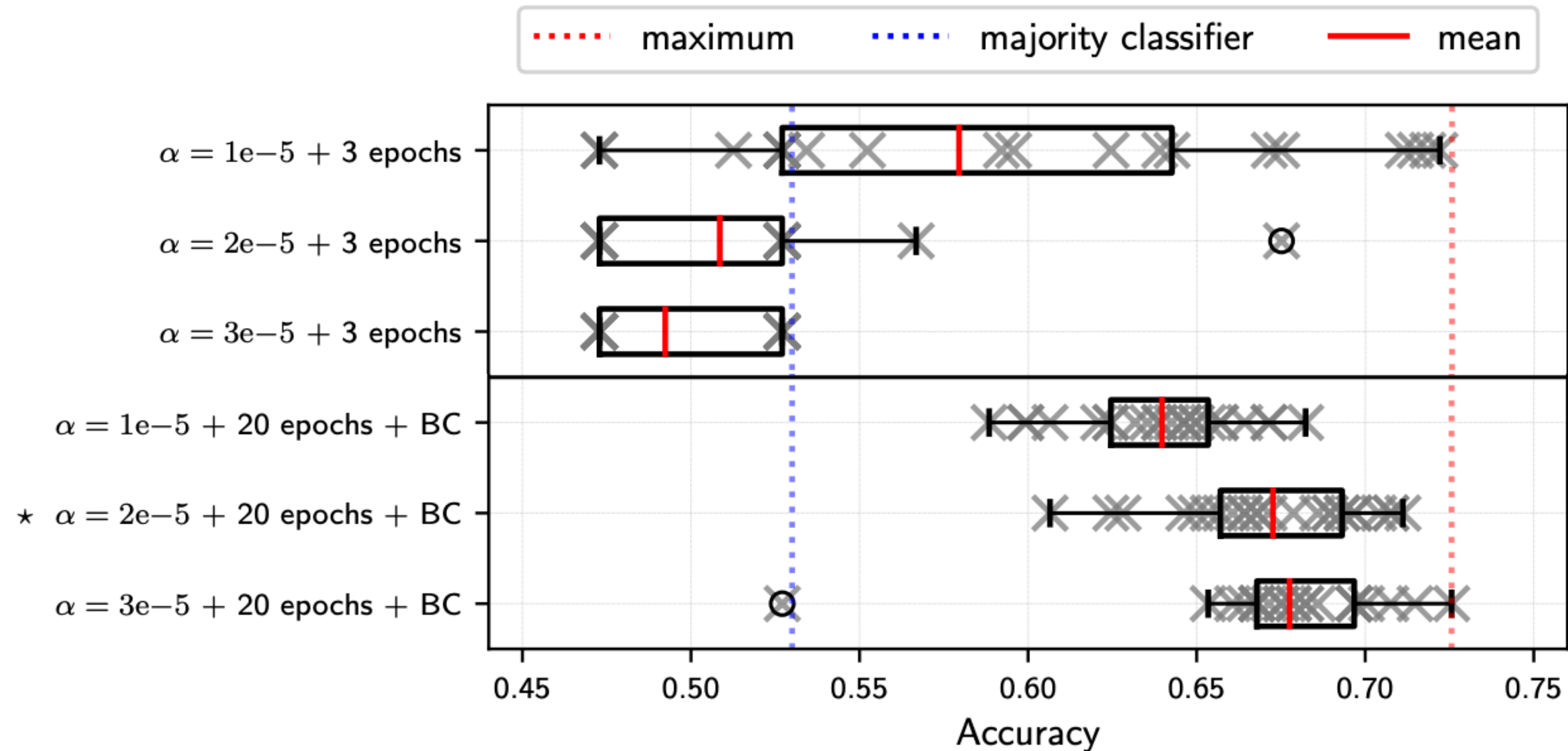
Guidelines for making comparisons

- Make sure you have baselines
 - Random
 - Majority Class
 - Conditioned Majority Class
- Other baselines
 - Simpler version of the model vs more complex
 - One layer FFNN
 - Unidirectional/Single-layer RNN
 - Retrieval baseline for generation problems
- Always consider: is the complexity warranted?
 - Different types of complexity
 - Simplicity in implementation vs simplicity in concept vs simplicity in model (number of parameters)
 - NN vs rule based vs NBs vs logistic regression

Classifier comparison

- Suppose you've assessed two classifier models. Their performance is probably different to some degree. What can be done to establish whether these models are different in any meaningful sense?
 - Practical differences
 - Confidence intervals
 - Wilcoxon signed-rank test (covered in the sentiment unit)
 - McNemar's test (covered in the sentiment unit)

Multiple runs with different seeds



On the stability of fine-tuning BERT: Misconceptions, explanations, and strong baselines
<https://arxiv.org/pdf/2006.04884v3.pdf> (Mosbach et al, ICLR 2021)

Tips for training and debugging

- TA Tutorial
- Stanford CS231n:
 - <https://cs231n.github.io/neural-networks-2/>
 - <https://cs231n.github.io/neural-networks-3/>
- CMU nn4nlp: Graham Neubig
 - <http://www.phontron.com/class/nn4nlp2021/schedule/debugging.html>
 - <https://www.youtube.com/watch?v=KRQHdwpfj-4>

Good luck on *your* project!